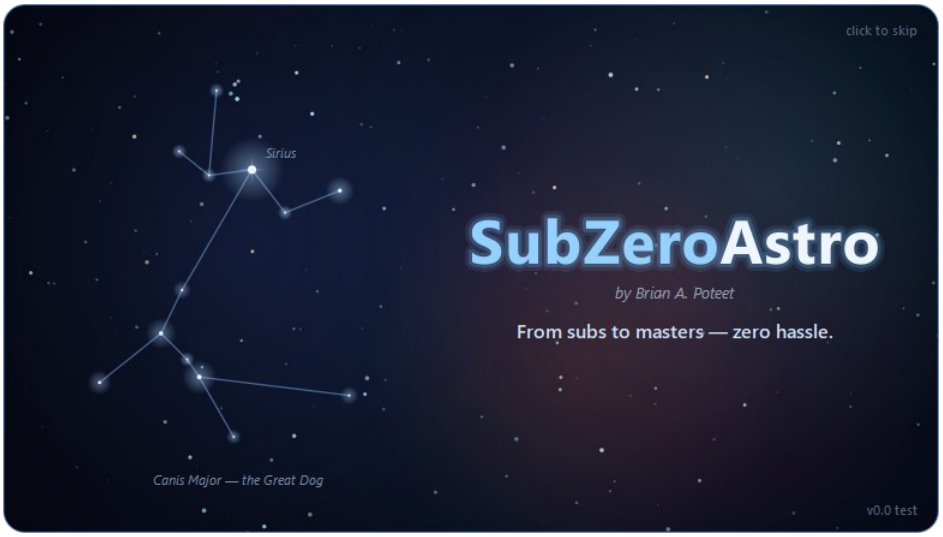
# SubZeroAstro

*From subs to masters – zero hassle.*

## User Manual

A beginner's guide to culling, stacking, processing and finishing your astrophotos




Version 0.9.45 beta · Windows and Mac (Apple silicon)

by Brian A. Poteet · October 2026

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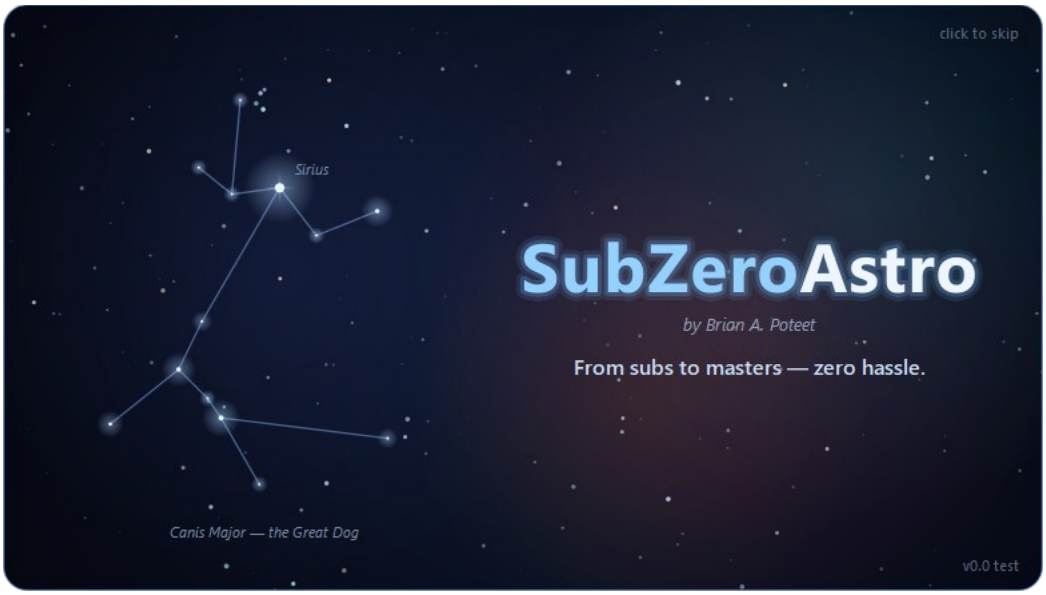
# 1 Welcome to SubZeroAstro

## 1.1 What SubZeroAstro does

SubZeroAstro takes you **from a folder of raw sub-frames to a finished astrophoto** - with as little hassle as possible. It walks you through four pages, one after the other, like turning the pages of a book:

| Page | What happens there |
|---|---|
| 1. Astro Image Quality and Cull | Every sub-frame is measured (star size, roundness, signal, cloud). You keep the good ones and set aside the bad ones. |
| 2. Calibrate and Stack | Your kept frames are calibrated with darks, flats and bias and combined into one clean **master** per filter. |
| 3. Astro Image Processing | The masters are cropped, plate solved, cleared of gradients and (optionally) sharpened, de-noised and separated from their stars with the RC-Astro XTerminators. |
| 4. Finishing Touches | A guided, pick-the-one-you-like workflow turns the masters into a finished colour picture you can save, sign and share. |

At every page you can also **stop and continue in PixInsight** - SubZeroAstro saves exactly what you have so far, plus a note of what to do next there.




*The SubZeroAstro start-up screen*

## 1.2 Who this manual is for

This manual is written for **newcomers** to astrophotography processing. No knowledge of PixInsight or other processing software is needed. Experienced imagers will find the PixInsight equivalent of each step mentioned along the way.

## 1.3 Words you will meet

| Word | What it means |
|---|---|
| Sub-frame (sub, light) | One single exposure of your target, e.g. 180 seconds. |
| Calibration frames | **Darks** (lens cap on, same exposure), **flats** (evenly lit, shows dust and vignetting), **bias** (shortest exposure) and **dark-flats**. They remove the camera's and telescope's own fingerprints. |
| Master | Many frames combined into one: a master light (your target) or a master dark / flat / bias. |
| Linear / stretched | Straight from the camera a picture is **linear** and looks almost black. **Stretching** (Brighten the Image) makes the faint detail visible. |
| OSC (colour camera) | One-shot-colour camera: records red, green and blue at once (e.g. ZWO ASI2600MC). |
| Mono camera | Records one filter at a time (L, R, G, B, Ha, OIII, SII); the colours are combined later. |
| Narrowband / dual-band | Filters that pass only the glow of hydrogen (Ha), oxygen (OIII) or sulphur (SII). A **dual-band** filter (e.g. Optolong L-Ultimate, Askar D1) passes two of them on a colour camera. |
| Plate solving | Working out exactly where in the sky the picture points, by matching its stars to a star catalogue. |
| FWHM | Star size (full width at half maximum). Smaller = sharper. |
| Eccentricity | How round stars are: 0 = round, 1 = a line. Above about 0.5 stars look elongated. |
| SNR | Signal-to-noise ratio: how clearly the stars rise above the sky. Drops with cloud, haze or moonlight. |
| Starless / stars-only | The picture split in two by StarXTerminator, so the nebula and the stars can be processed separately. |

# 2 What you need

## 2.1 Computer requirements

SubZeroAstro runs on **Windows** and on **Macs with Apple silicon**. Astro images are big (a 26-megapixel camera makes ~50 MB files; a night can easily be 20-50 GB), so more memory and a fast disk make everything quicker.

| | Minimum | Recommended |
|---|---|---|
| Operating system | Windows 10 (64-bit) or a Mac with Apple silicon (M1 or newer) and a recent macOS | Windows 11, or macOS 14 (Sonoma) or newer |
| Processor | 4 cores | 8 cores or more |
| Memory (RAM) | 16 GB | 32 GB or more (64 GB for 60-megapixel cameras or very large stacks) |
| Free disk space | 20 GB plus room for your images | An SSD with 100 GB or more free - stacking keeps temporary aligned frames on the local disk |
| Graphics card | Not required | An NVIDIA RTX card (Windows) or Apple silicon GPU makes the RC-Astro XTerminators many times faster |
| Screen | 1600 × 900 | 1920 × 1080 or larger |

> **Tip:** Images on a network drive (NAS) work fine. SubZeroAstro reads them carefully (3 files at a time) and remembers the measurements, so re-opening a session is quick.

## 2.2 Companion programs (optional but recommended)

SubZeroAstro works on its own, but it can call these free or commercial programs for the best results. Install them **before** the first setup (Chapter 4); SubZeroAstro finds them automatically in their usual places. Each time it starts, page 1 checks for them quietly in the background and shows an amber note if one is missing or not licensed (4.1).

| Program | Used for | Where to get it | Without it |
|---|---|---|---|
| **ASTAP** + a star database (D50 suits most setups) | Plate solving (page 3) | hnsky.org/astap.htm (free) | Plate solving is skipped (PixInsight can do it later) |
| **GraXpert** | AI gradient removal (page 3) | graxpert.com (free) | A built-in gradient remover is used |
| **RC-Astro CLI** with your BlurXTerminator, NoiseXTerminator and StarXTerminator licences | Sharpening, noise reduction and star removal (pages 3 and 4) | Your RC-Astro account (rc-astro.com) | These steps are skipped; Take Stars Out is not available |

| Program | Used for | Where to get it | Without it |
|---|---|---|---|
| **PixInsight** (optional) | Further processing after any page | pixinsight.com | Not needed - SubZeroAstro can finish the picture |

## 2.3 Your images

- SubZeroAstro reads **FITS** (`.fit`, `.fits`, `.fts`) and PixInsight **XISF** (`.xisf`) files on every page - light frames, calibration frames and masters - from capture programs such as N.I.N.A., ASIAIR, SharpCap and smart telescopes (ZWO Seestar, DWARF, Vaonis Vespera, Stellina, eVscope). XISF files can be uncompressed or compressed with any of PixInsight's options (zlib, LZ4, LZ4-HC, Zstandard, with or without byte shuffling).
- Keep each night's lights in their own folder, and your calibration frames in a **calibration library** folder with one subfolder per telescope (see 4.3).
- Your original files are **never changed or deleted**. Rejected frames are only moved into a `REJECTED` subfolder.

# 3 Installing SubZeroAstro

## 3.1 Windows

1. Open the SubZeroAstro releases page: **github.com/bapoteet/SubZeroAstro/releases**.
2. Under the newest release, click **SubZeroAstro-Windows.zip** to download it.
3. Right-click the downloaded zip › **Extract All…** and choose a permanent place, e.g. `C:\Astro\SubZeroAstro`.
4. Open the extracted folder and double-click **SubZeroAstro.exe**. (Right-click › *Pin to Start* or *Send to › Desktop* gives you a shortcut.)
5. If Windows shows **"Windows protected your PC"**, click **More info › Run anyway**. This appears because the app is new and not yet code-signed; it is safe to continue.

## 3.2 Mac (Apple silicon)

1. Open **github.com/bapoteet/SubZeroAstro/releases** and download **SubZeroAstro-macOS-AppleSilicon.zip** from the newest release.
2. Double-click the zip in your Downloads folder; **SubZeroAstro.app** appears.
3. Drag **SubZeroAstro.app** into your **Applications** folder.
4. The first time only: **right-click** (or Control-click) the app › **Open**, then **Open** again in the message. After that it opens normally.
5. If macOS says the app "is damaged" or "can't be opened", open **Terminal** (Applications › Utilities) and type: `xattr -dr com.apple.quarantine /Applications/SubZeroAstro.app` then press Return and open the app again.

> **Why the warnings?:** Both messages appear because SubZeroAstro is not yet signed by Microsoft or notarized by Apple. Signing is being set up for future versions.
>
> **Mac: window size:** SubZeroAstro opens filling the screen, with the menu bar and Dock still showing. Avoid the green button's full-screen mode on a Mac: it hides the menu bar (where **Stack › Build Calibration Masters…** and **Help** live). To fill the screen yourself, **Option-click** the green button, or double-click the window's title bar. If you make the window smaller, it opens at that size next time.

## 3.3 Updating to a new version

Download the newest zip from the releases page and replace the old app folder (Windows) or the app in Applications (Mac). **Your settings, tool locations, rig names, saved measurements and saved progress are kept** - they are stored separately from the app.

## 3.4 Removing SubZeroAstro

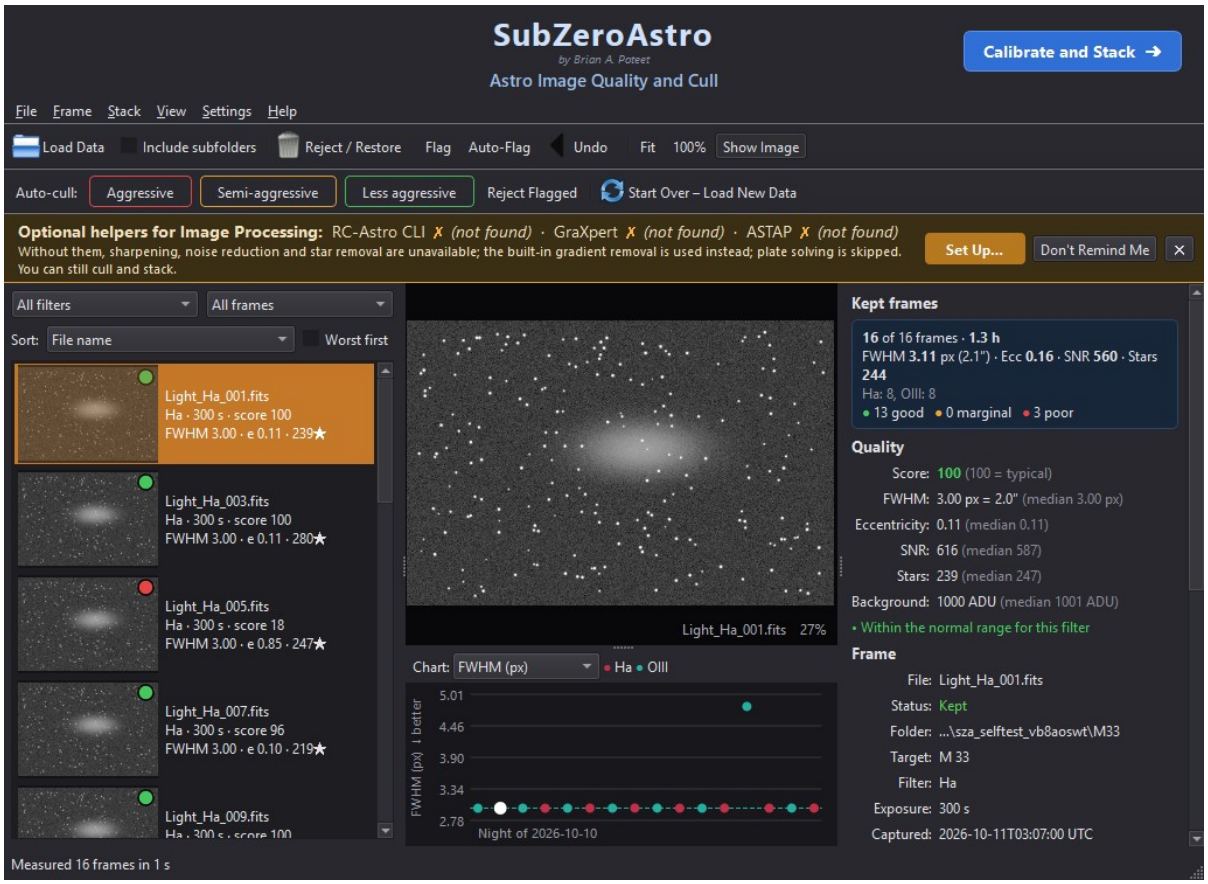
Delete the app folder (Windows) or drag SubZeroAstro from Applications to the Bin (Mac). Saved progress can be cleared first under **Settings › Saved Progress**. Your images are never touched.

# 4 First-time setup

## 4.1 Tell SubZeroAstro where your tools are

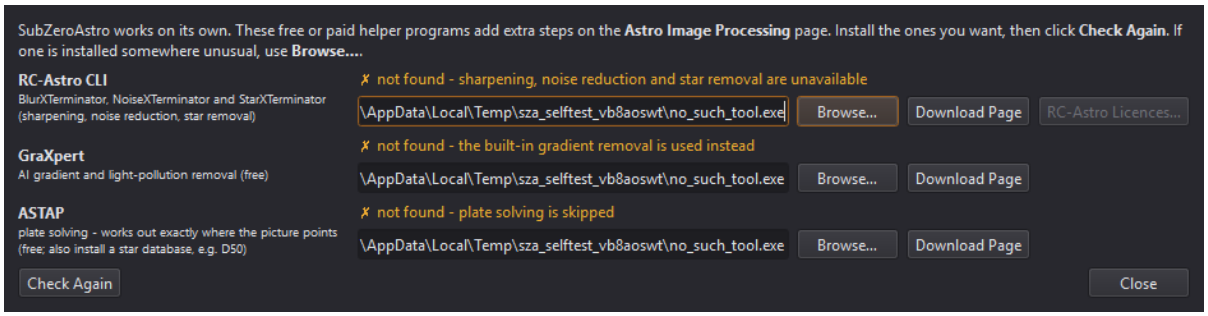
When SubZeroAstro starts, it checks for ASTAP, GraXpert and the RC-Astro CLI in the background. If one is missing - or RC-Astro is installed but a tool is not licensed - a slim **amber banner** appears at the top of page 1, saying what you would miss (for example *plate solving is skipped*). Nothing is wrong: you can still cull and stack. The banner has two buttons:

- **Set Up…** - opens the **Your Tools** window (also in **Settings › Your Tools (ASTAP, GraXpert, RC-Astro)…**).
- **Don't Remind Me** - hides the banner until something else changes. Use it if you have chosen not to use one of the programs.




*The amber banner on page 1 when helper programs are missing (example)*

The **Your Tools** window lists each program with its status, a **Download Page** button for anything missing, and **Browse…** to point at a program installed somewhere unusual. Click **Check Again** after installing something.




*Settings › Your Tools: status, Browse… and Download Page for each helper program*

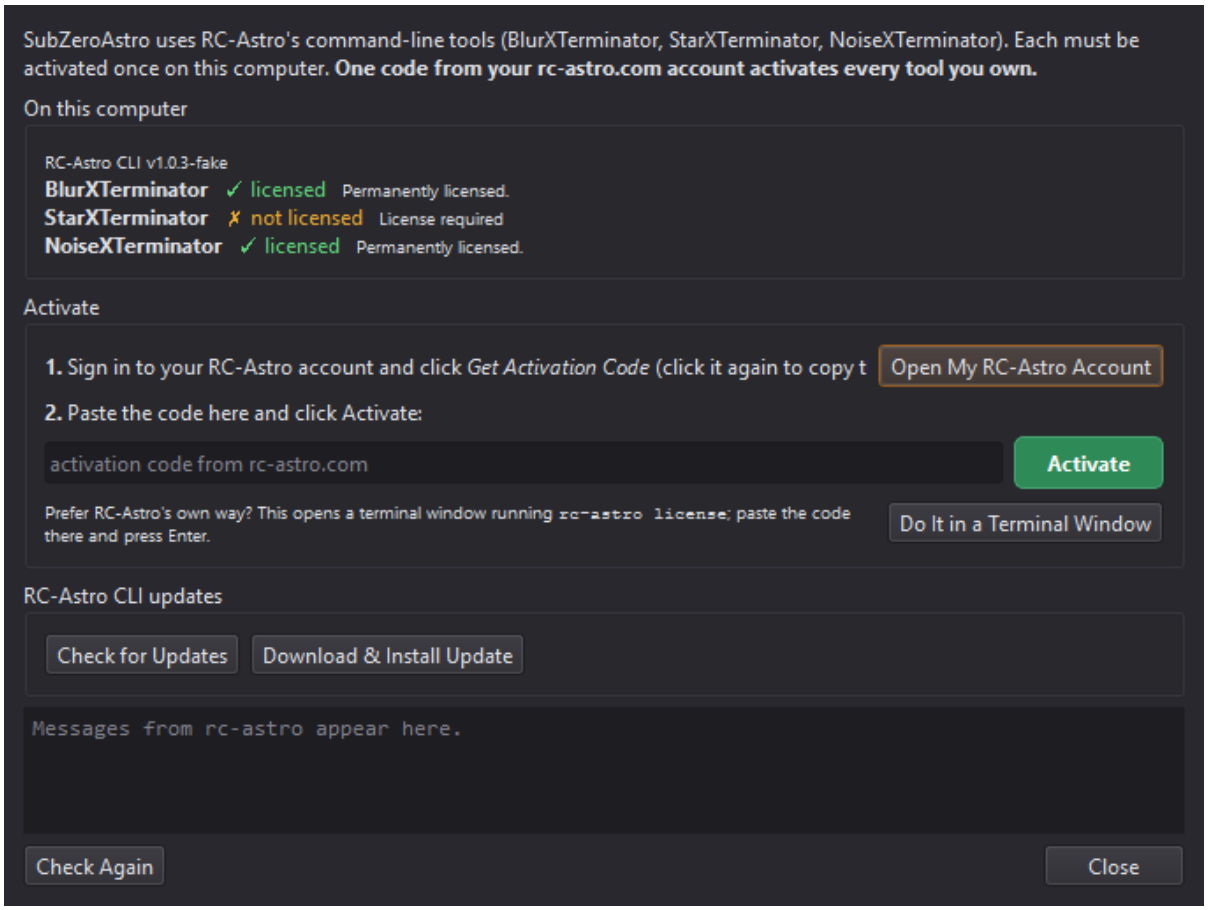
The same programs are also shown in the **Your tools** box at the top of page 3 (**Astro Image Processing**), with a green ✔ **found** when detected. If one is missing, click **Browse…** next to it and pick the program:

| Tool | Typical location on Windows | Typical location on a Mac |
|---|---|---|
| ASTAP | `C:\Program Files\astap\astap.exe` | `/Applications/ASTAP.app` (or `astap_cli`) |
| GraXpert | `C:\Users\<you>\AppData\Local\Programs\GraXpert\GraXpert.exe` | `/Applications/GraXpert.app` |
| RC-Astro CLI | `C:\Program Files\RC-Astro\CLI\rc-astro.exe` | where the RC-Astro installer put `rc-astro` |

**Activating RC-Astro licences.** You do not need a terminal: click **RC-Astro Licences…** (next to the RC-Astro CLI row on page 3, or in the Your Tools window). The window shows which tools are licensed on this computer. To activate:

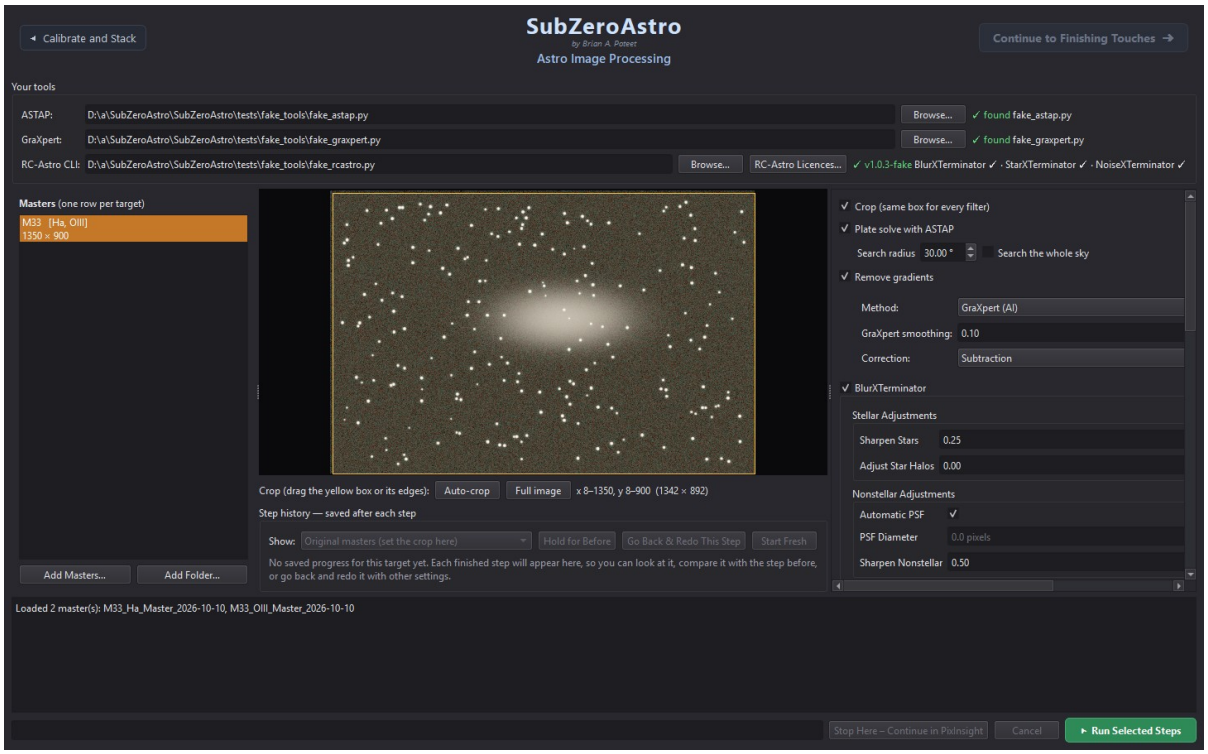
1. Click **Open My RC-Astro Account** and sign in on rc-astro.com.
2. Click **Get Activation Code** on the account page, then click it again to copy the code.
3. Paste the code into the box and click **Activate**. **One code activates every RC-Astro tool you own.**

The tools turn green and page 3 is updated straight away. If activation does not work, **Do It in a Terminal Window** runs RC-Astro's own `rc-astro license` command, where you paste the code and press Enter. **Check for Updates** and **Download & Install Update** keep the RC-Astro CLI itself up to date.




*The RC-Astro Licences window (example: StarXTerminator not yet activated)*

For the RC-Astro CLI the status line lists **BlurXTerminator ✓ · StarXTerminator ✓ · NoiseXTerminator ✓** for every product licensed on this computer. Below the XTerminator settings, **RC-Astro device** chooses the processor: pick your dedicated graphics card (e.g. *NVIDIA RTX…*) - it is much faster than integrated graphics.




*Page 3 with the tools box at the top (example data)*

## 4.2 Choose where results are saved

On page 2 (**Calibrate and Stack**) set two folders under **Save locations** once; they are remembered:

- **Calibration masters** - your calibration library (4.3). New master darks, flats and bias are saved there and found again automatically next time.
- **Master lights** - where your stacked masters go, one subfolder per target, e.g. `…\Master Lights\IC1805\IC1805_L-Ultimate_Master_2026-10-09_to_2026-10-11.fits`.

## 4.3 The calibration library

SubZeroAstro keeps calibration masters in a tidy structure inside the calibration folder:

`<calibration folder>\<telescope rig>\Masters\<camera>\Bias | Darks | Flats\<filter>\…`

- **Rig names**: the first time, pick a rig name for each camera + focal length (e.g. *265mm scope* or *SQA55*). This matters for ASIAIR, which records the mount, not the telescope.
- **Darks** are matched by exposure, gain, offset and temperature (±3 °C). **Flats** are matched by camera, filter and rig, nearest in date. **Bias** or **dark-flats** are needed to calibrate flats properly.
- You can add a folder of raw calibration sub-frames; SubZeroAstro builds the masters and saves them into the library.

## 4.4 Settings worth knowing

- **Settings › Star measurement** - *Fast* (default, about 3-4× quicker on a network drive), *Full frame* or *Off*.
- **Settings › Your Tools (ASTAP, GraXpert, RC-Astro)…** - see, locate and download the helper programs, and activate RC-Astro licences (4.1).

- **Settings › Saved Progress…** - where crash-recovery copies are kept, how much space they use and how long they are kept (30 days by default).
- **Settings › Show Intro Animation at Startup** - turn the start-up animation on or off. *Help › Play the Intro Animation* plays it any time.

# 5 Finding your way around

## 5.1 The banner and page buttons

Every page has the same banner at the top: **SubZeroAstro**, the author and the **version** in small print (e.g. *v0.9.10 beta*), and the **page title**. On the right is a blue **Continue to …** ➔ button that turns to the next page; on the left a grey ◀️ **previous page** button goes back. Going back keeps your work: press Continue again and the page you left comes back exactly as it was.

## 5.2 The log and the status bar

The text area at the bottom of each page is the **log**: it tells you what was done, what was found and any warnings, in plain language. The orange bar shows progress and an estimate of the time left.

When a page has finished its job, its green run button turns **amber** - ✔ **Stacked – Run Again?** on page 2, ✔ **Done – Run Again?** on page 3 - so you can see at a glance that the page is done. Pressing it again asks first, so nothing is redone by accident. Add new images and it turns green again.

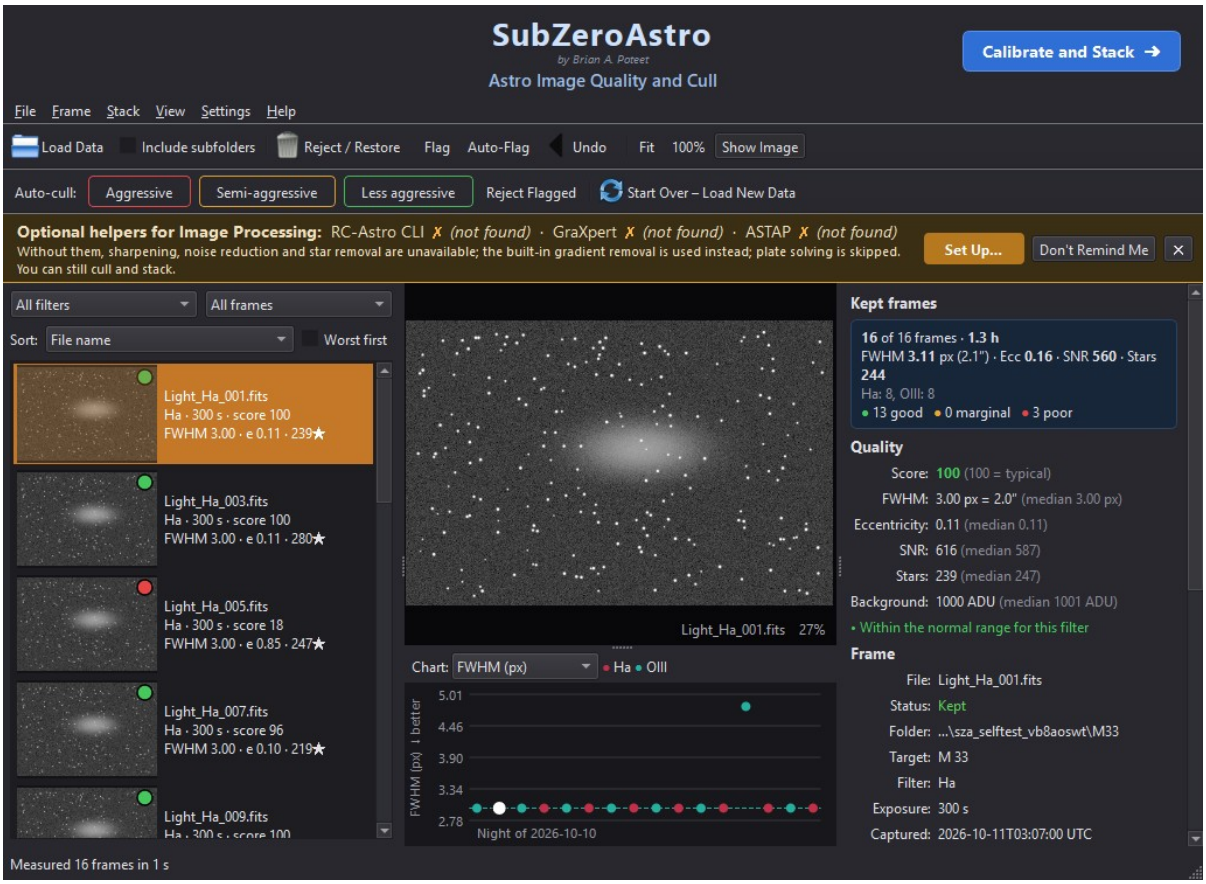
## 5.3 Saved progress and undo

- **Page 1**: your folders and flags are saved as you go. **Ctrl+Z** undoes rejects, flags and auto-cull.
- **Pick up where you left off**: when you start SubZeroAstro again - after quitting normally or after a crash - it offers **Pick Up Where I Left Off**: your frames and flags come back and it goes straight to the page you were on (Calibrate and Stack, Image Processing with the same masters, or Finishing Touches, where your steps resume). **Start Fresh** begins a new project; after *End this Project and Start New Project* nothing is offered.
- **Page 2**: each master is noted as soon as it is saved; an interrupted run offers **Continue With the Rest**.
- **Page 3**: the image is saved after every step; **Go Back & Redo This Step** and **Start Fresh** are under the preview.
- **Page 4**: every step is saved: **Undo** (Ctrl+Z), **Go Back to Here**, **Back to Start**, and reopening the same masters later resumes the project.

## 5.4 Stop Here – Continue in PixInsight

The purple **Stop Here – Continue in PixInsight** button (pages 2, 3 and 4) saves the image exactly as it is right now as PixInsight-ready XISF, writes a short note listing the PixInsight processes to use next, and opens the folder with the files selected.

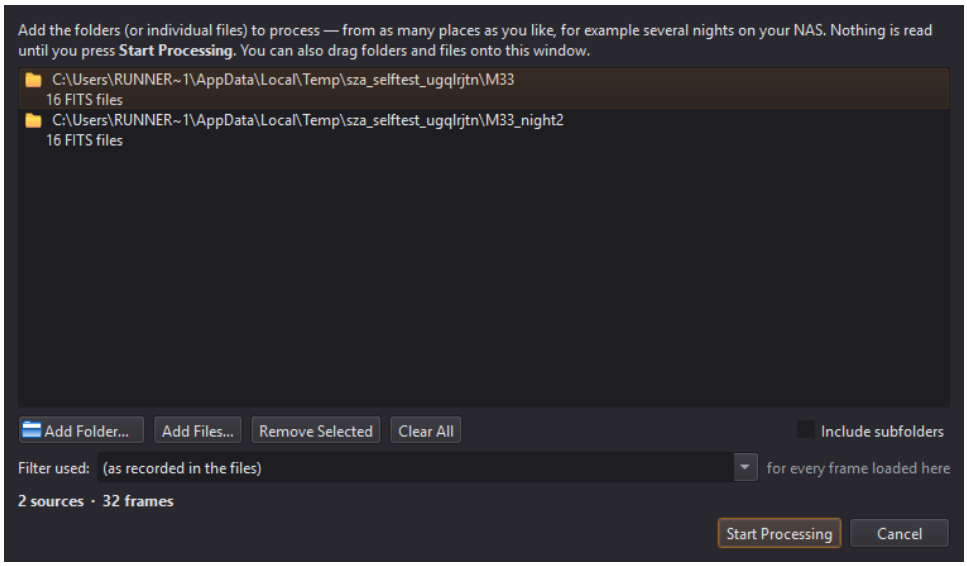
# 6 Page 1 – Astro Image Quality and Cull




*Page 1: frame list (left), the selected frame, the session chart and the measurements (example data)*

## 6.1 Loading your sub-frames

1. Click **Load Light Frame Data** (Ctrl+O), then **Add Folders…**. You can pick **several folders at once**: click one, then **Shift**-click to add a range, or **Ctrl**-click (**Cmd**-click on a Mac) to add others one by one. Click **Add Folders…** again for folders in other places (other nights, other drives), or **Add Files…** for individual files.
2. **Include subfolders** is ticked by default, so nights kept in subfolders are found too (FLAT, DARK, BIAS, CALIB and OFFSET folders are skipped). Untick it to load only the files directly in each folder; your choice is remembered.
3. If your capture software does not record the filter (e.g. ASIAIR with a manual filter drawer), choose it under **Filter used**.
4. Press **Start Processing**. Frames appear at once and are measured in the background.




*The Load Light Frame Data window*

> **Calibration frames?:** Load Light Frame Data is for **light frames only**. To turn bias, darks and flats into masters, use **Stack › Build Calibration Masters…** instead (7.6) - no light frames needed.
>
> **Tip:** If the file names mention the filter (e.g. `…_SQA55Lult_0001.fit`), SubZeroAstro notices after the first few frames and offers to use it for the whole set. Duplicate files are skipped and reported.

## 6.2 Reading the measurements

Each frame gets a coloured **badge** (green = good, amber = check it, red = probably reject) and a **score** where 100 = a typical frame for that filter. Frames are only ever compared with frames from the **same camera and filter**.

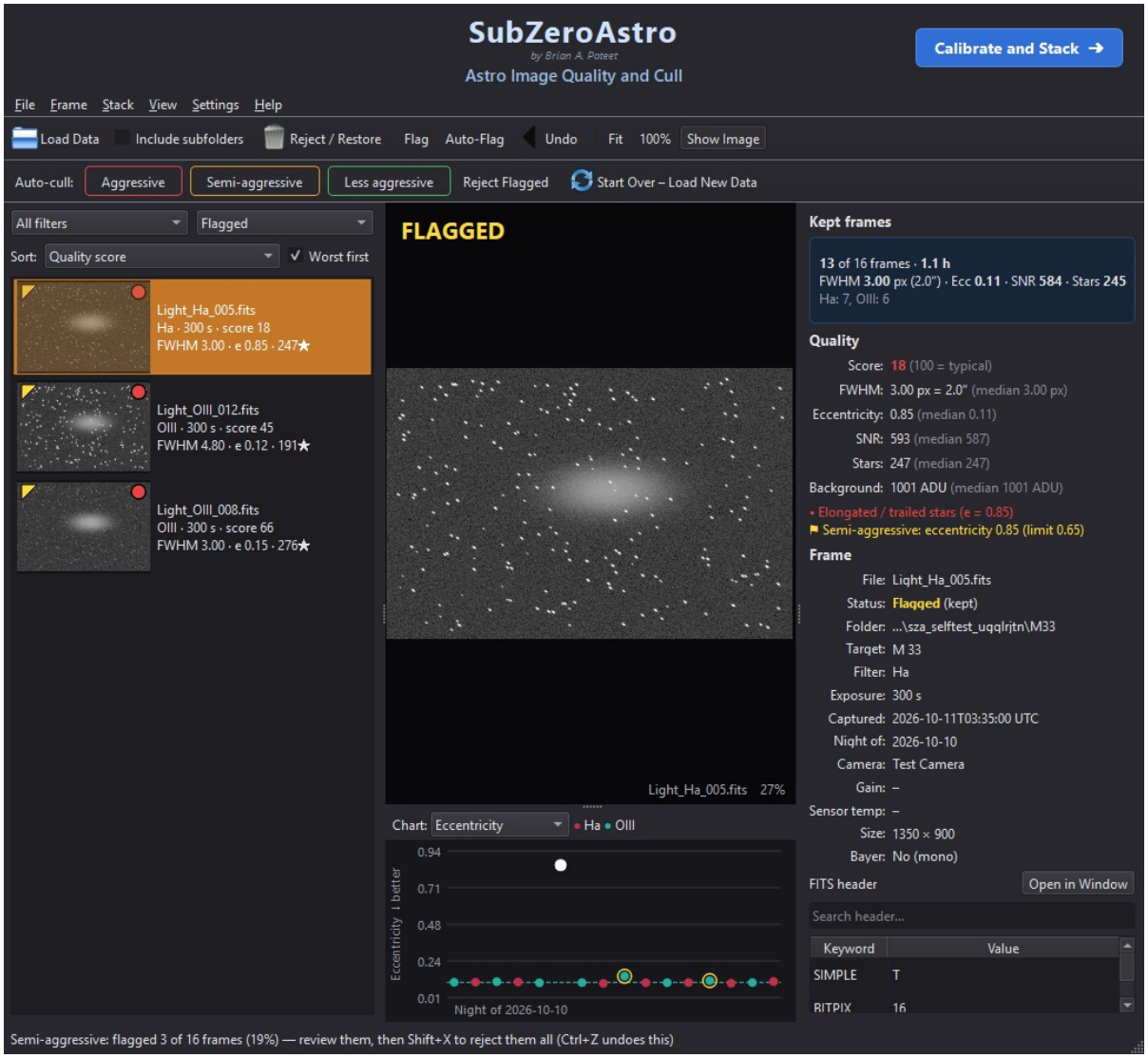
| Measurement | What it tells you | Bad sign |
|---|---|---|
| FWHM | Star size in pixels (and arcseconds) | Much larger than the median: poor focus or seeing |
| Eccentricity | Star roundness (0 = round) | Above ~0.5: tracking, guiding or wind |
| SNR | How clearly stars stand out | A drop: cloud, haze, dew, moonlight |
| Stars | Number of stars detected | A sudden drop: cloud or trees |
| Background | Sky brightness | A rise: moon, twilight, light pollution |

The **session chart** below the image shows one dot per frame in capture order. Choose what to plot, **click** a dot to jump to that frame, or **drag** across the chart to select a range. The **Kept frames** box on the right shows the averages of what you are keeping.

## 6.3 Choosing which frames to keep

1. Click an **Auto-cull** button - **Aggressive**, **Semi-aggressive** or **Less aggressive**. It only **flags** frames (yellow ring); hover over a button first to see how many it would flag.
2. Review the flagged frames (switch the list to *Flagged*); the measurements panel says why each was picked. Press **F** to un-flag one you want to keep.

3. Press **Reject Flagged** (Shift+X). Rejected frames are moved into a `REJECTED` subfolder - nothing is deleted, and **Ctrl+Z** brings them back.




*After Semi-aggressive: suspect frames are flagged for review (example data)*

**Auto-Flag** (Ctrl+F) lets you set your own limits. **X** rejects or restores the selected frame(s), **F** flags them, **D** clears all flags.

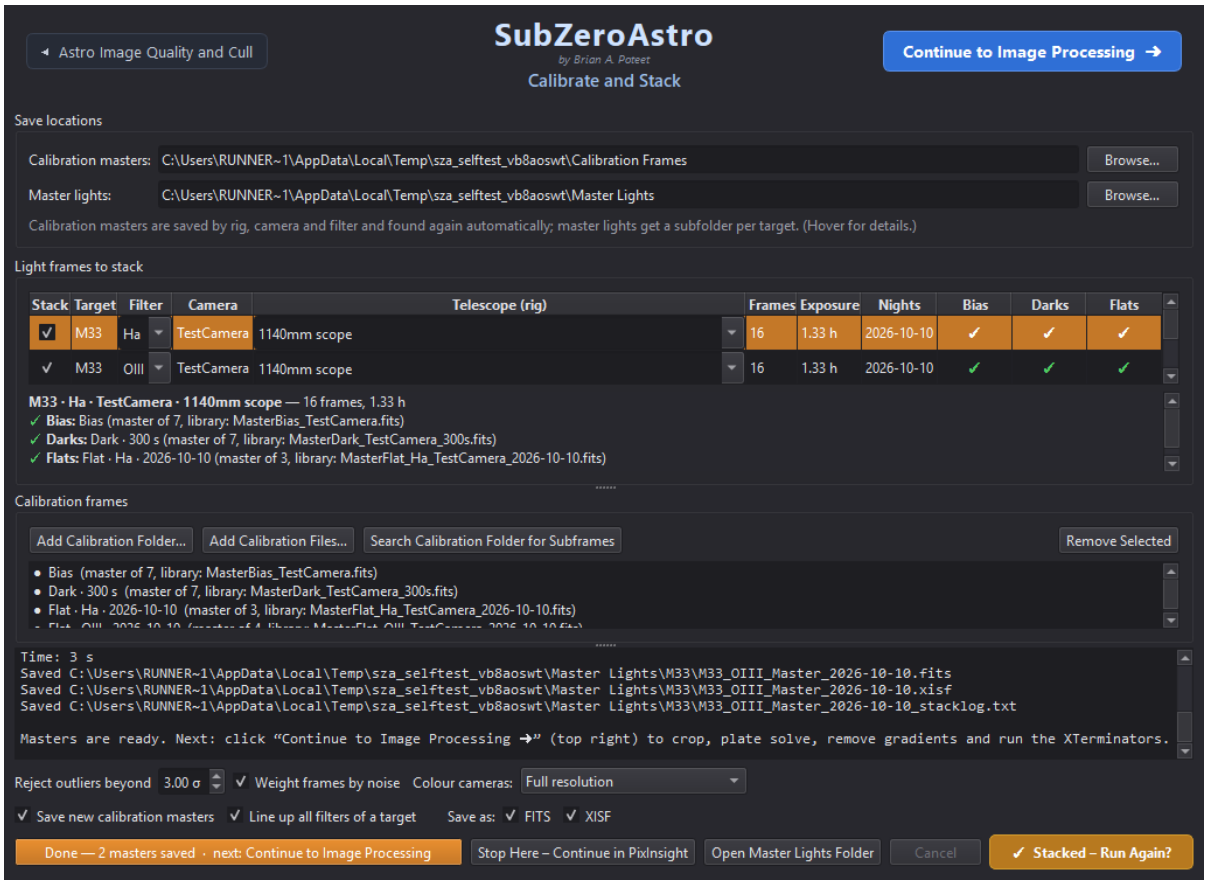
## 6.4 Filters and the FITS header

- **Frame › Set Filter Name…** changes the filter for the selected frames (your files are not changed). Use it when the filter is missing or wrong - flats are matched by filter.
- **Ctrl+H** opens the full **FITS header** of the selected file, with search and Copy All.
- **Ctrl+I** hides or shows the image - useful on a slow network drive.

## 6.5 On to stacking

When you are happy, click **Calibrate and Stack ➜** at the top right. Only your kept frames go to page 2.

# 7 Page 2 – Calibrate and Stack




*Page 2 after stacking: one row per target and filter, with the calibration found for each (example data)*

## 7.1 The light frame groups

Your kept frames are grouped by **target + filter + camera + telescope** - one row each, with the number of frames, total exposure and nights. The **Filter** of a row can be changed from its drop-down (for example to match your flats).

## 7.2 Calibration: what the ✓ ⚠ ✗ mean

| Mark | Meaning |
|---|---|
| ✓ green | A matching calibration master was found (or will be built). |
| ⚠ orange | Usable, with a note - e.g. darks scaled from another exposure, or flats taken long ago (*"dust may have moved"*). It **will** be used. |
| ✗ red | Nothing suitable - e.g. *No flats for filter OIII*. You can still stack, without that correction. |

Click a row to read the details below the table. Add calibration with **Add Calibration Folder…**, **Add Calibration Files…** or **Search Calibration Folder for Subframes**.

**No flats at all? SkyFlats (Mote Buster).** When a filter has no flats, SubZeroAstro can make a flat from your **light frames** (option *SkyFlats (Mote Buster) when flats are missing*, on by default; at least

12 frames). It uses every light with that filter, camera and rig - from all your targets and nights, because the stars land somewhere different in each. Each frame is scaled to the same brightness, its stars are hidden, and its own light-pollution tilt is taken out (that tilt changes from frame to frame, while the vignetting stays put). Of every pixel's values only the lower half is used - a nebula can only add light - and what remains is what your optics and dust do to an even sky: a flat. The Flats cell then shows ⚠️ *SkyFlat (Mote Buster) (made from N light frames)*. SkyFlats are never saved to your library. (For donuts that remain, *Mote Buster – Fix Dust Donuts* on Finishing Touches works on one picture.)

**Every SkyFlat is checked before it is used.** If it may not be good enough - too few usable frames, grainy, a brightness range no vignetting has (nebulosity or a light-pollution gradient got in), large areas filled in - or it cannot be made at all because the master dark or bias is as bright as the sky (a light leak, or the wrong exposure or gain), SubZeroAstro stops and asks: **Proceed Without Flats** (the default), **Use the SkyFlat Anyway**, or **Cancel Stacking**. The reasons are listed in the box and in the log. A poor flat adds blotches or rings instead of removing them; without flats you can still fix dust donuts afterwards with *Mote Buster – Fix Dust Donuts* on Finishing Touches.

> **SkyFlats:** A SkyFlat is best from several targets. From a single target whose nebula fills most of the picture, part of the nebula could be mistaken for vignetting - the Flats note warns you. Real flats are still the best: take some when you can.

**Calibration check.** Every stack checks its dark (or bias) against the lights and writes a *Calibration check* line in the log: light background - dark = sky left. If a master does not fit - **brighter than the lights**, **uneven like a picture** (light leaked in while the darks were taken: lens cap, filter wheel, focuser or room light) or **practically zero** (saved on another scale) - a window says so when stacking finishes. Such a frame makes bright bands, borders or blotches that no later step can fix: remove it from the *Calibration frames* list, or take new ones (scope fully covered, same gain, offset, temperature and exposure as the lights).

> **Important:** If the Flats cell says the flats have **no matching bias or dark-flats**, the correction will be too strong: bright corners and dark rings. Add a bias or dark-flats for that camera and gain.

## 7.3 Stacking options

- **Reject outliers beyond 3.0 σ** - removes satellite trails, planes and hot pixels. The default suits almost everything.
- **Weight frames by noise** - better frames count more.
- **Colour cameras: Full resolution** (default) or half resolution (faster, less noise per pixel).
- **Save as** FITS and/or XISF; **Save new calibration masters** into your library.
- **Line up all filters of a target** (on) - all masters of a target share one pixel grid, ready to combine (LRGB, SHO).

**Sharp stars to the corners.** Every telescope bends the field a little, mostly at the edges. After a **meridian flip** (or between nights with another setup) that bending is turned round with the camera, so a plain shift-and-rotate alignment lines up the centre but leaves stars a pixel or two apart near the edges - doubled or smeared corner stars. SubZeroAstro measures each frame's stars against the reference and, when the edges are off, lines that frame up with a smooth distortion model instead. The stack log says how many frames were corrected.

## 7.4 Running the stack

Press ▶️ **Check & Stack**. Anything missing is listed first, with **Stack Anyway** or **Add Calibration Frames First…**. The log shows each step. If a run is interrupted, the next run offers **Continue With the Rest**. When it finishes, the button turns amber: ✔ **Stacked – Run Again?**. Stacking again never overwrites a master - new versions are saved as _2, _3…

## 7.5 Results

Masters are saved in your master lights folder, one subfolder per target, with a `_stacklog.txt` listing every frame used. **Open Master Lights Folder** shows them. Click **Continue to Image Processing ➔** to carry on.

## 7.6 Build Calibration Masters (no lights needed)

Took a fresh set of darks, flats or bias? Turn them into masters right away - **no light frames needed**: on page 1 choose **Stack › Build Calibration Masters…** (or click **Build Calibration Masters…** in the *Calibration frames* box on page 2). Choose the folder with the subframes - bias, darks, flats and dark-flats can all be mixed together in one folder (for example *RedCat51Calis*), or sit in subfolders. SubZeroAstro reads each file's header (and, if the header doesn't say, the file or folder name: *Darks*, *Flats*, *Bias*…) and sorts them:

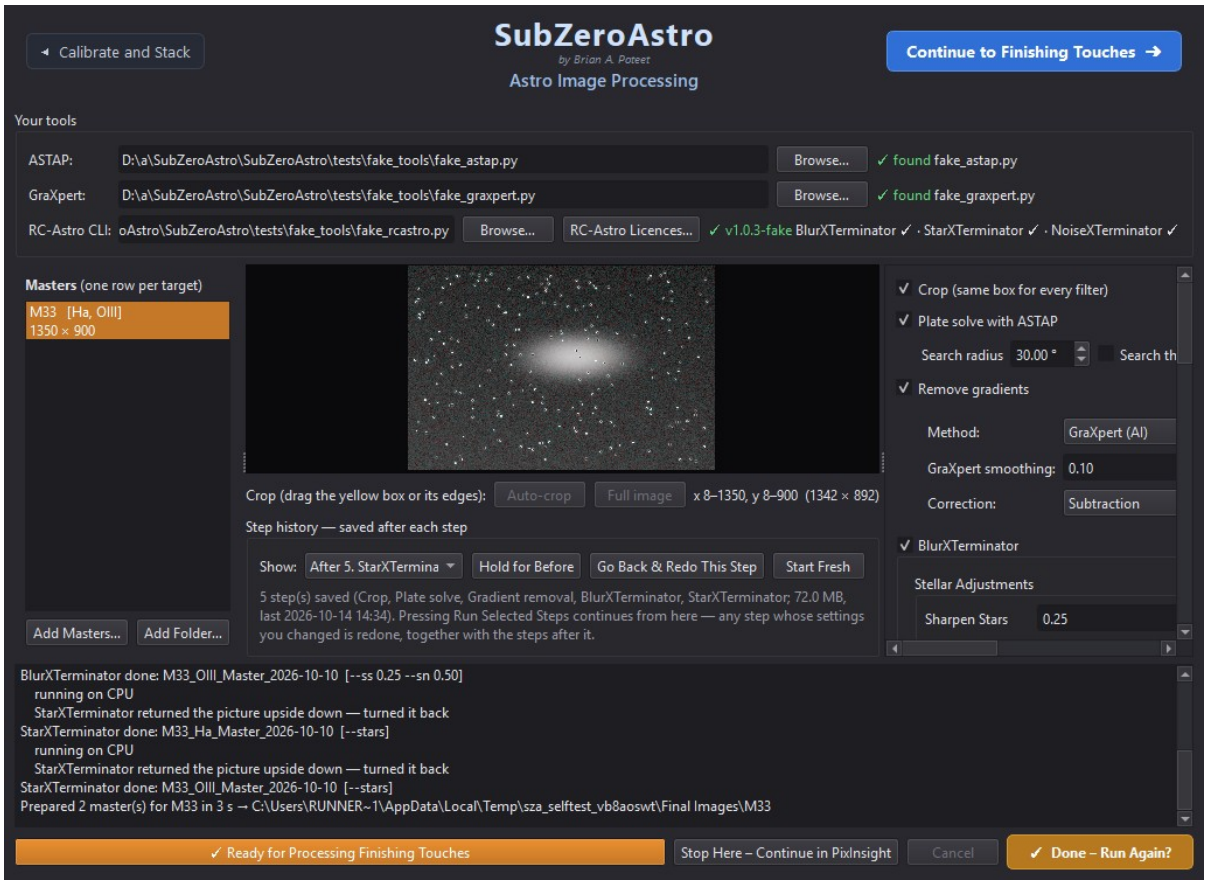
- **Bias** - one master per gain and offset.
- **Darks** - one master per exposure, gain, offset and temperature (180 s and 300 s darks make two master darks).
- **Flats** - one master per filter (and night). Each flat is calibrated with dark-flats or short darks of the same exposure, or else a bias - from the same folder or already in your library.
- **Dark-flats** - saved as masters too, for next time.

The telescope (rig) name is taken from the folder name (*RedCat51Calis → RedCat51*) or the headers - check it, because flats are only used for lights from the same rig. Untick any set you don't want, then click **Build Masters**. Each master is saved to your calibration library (*<library>/<rig>/Masters/<camera>/…*) with its settings in the name, and is used automatically whenever you stack lights with matching settings.

Every master is checked as it is built. You are warned when a dark or bias is **uneven like a picture** (light leaked in), a dark is **far brighter than the bias** (light, or the camera not cooled), a bias is **practically zero** (saved on a 0..1 scale or with a pedestal taken off), or flats are **over-exposed** (clipped) or **very dim**. A warned master is still saved, so you can compare - but better to retake that set.

> **Taking good darks:** Darks: cap the telescope (or take the camera off) and cover it, at night or in a dark room, with the filter wheel on a filter - not an empty slot. Match the lights' exposure, gain, offset and temperature.

# 8 Page 3 – Astro Image Processing




*Page 3 after processing: masters (left), preview with crop box, step history, and the step settings (right)*

## 8.1 Masters and target sets

The masters from page 2 are loaded automatically (or use **Add Masters…** / **Add Folder…**). Masters of the **same target** form one row - e.g. *NGC281 [Ha, OIII, SII]*. If they were stacked in separate runs the row says **will be lined up**: they are matched star-to-star onto one grid before anything else, and edges not covered by every filter are trimmed.

## 8.2 Crop

The yellow box on the preview is the crop. **Auto-crop** trims the ragged stacking edges; drag the box or its sides to adjust, or **Full image** to keep everything. The same box is used for **every filter**, so they stay aligned.

## 8.3 Plate solve

ASTAP finds where the picture points in the sky, using the position and pixel scale recorded in the masters. If it fails, tick **Blind solve** (slower, searches the whole sky).

## 8.4 Remove gradients

Removes light-pollution glow and uneven backgrounds. **GraXpert (AI)** is recommended; **Smoothing** 0.1 and **Correction: Subtraction** are good defaults. Without GraXpert a built-in model is used.

## 8.5 The XTerminators

| Tool | What it does | Good starting settings |
|---|---|---|
| BlurXTerminator | Sharpens stars and detail | Sharpen Stars 0.25, Sharpen Nonstellar 0.50, Automatic PSF on |
| NoiseXTerminator | Reduces noise (a light pass on unstretched data) | Denoise 0.6 |
| StarXTerminator | Separates a **starless** and a **stars-only** image | Generate Star Image on; **Unscreen Stars off** |

Hover over any setting for a plain-language explanation. **Unscreen Stars** should stay **off** here: the stars image is then simply *original minus starless* and is added back later (what Finishing Touches expects). Turn it on only if you will recombine stretched images with a screen blend in PixInsight.

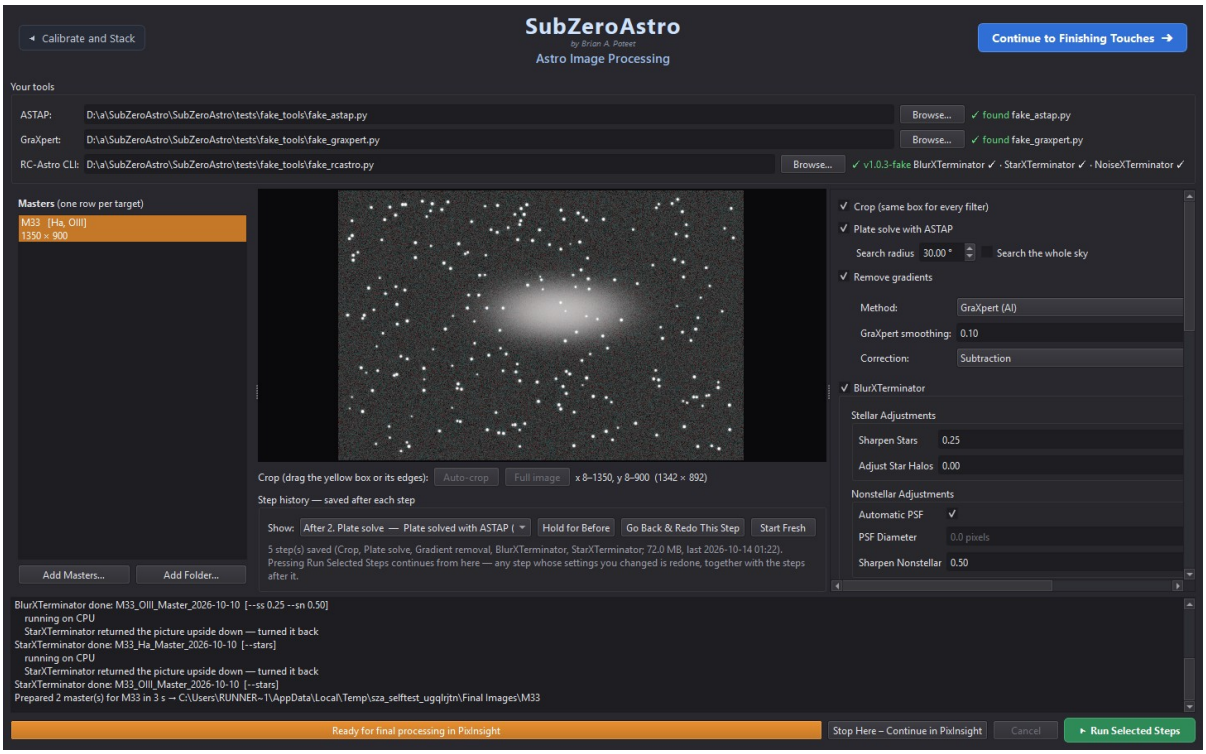
## 8.6 Saving the results

- The first time you run, SubZeroAstro asks **where the processed images should go**: *Next to the Masters* (a `Prepared` folder), *Folder Used Last Time*, or *Choose a Folder…*. Each target gets its own subfolder.
- **Linear masters** as XISF and/or FITS (these are what you process further), and a **stretched preview** as TIFF, JPG and/or PNG in the chosen **Preview palette**.

## 8.7 Running and the step history

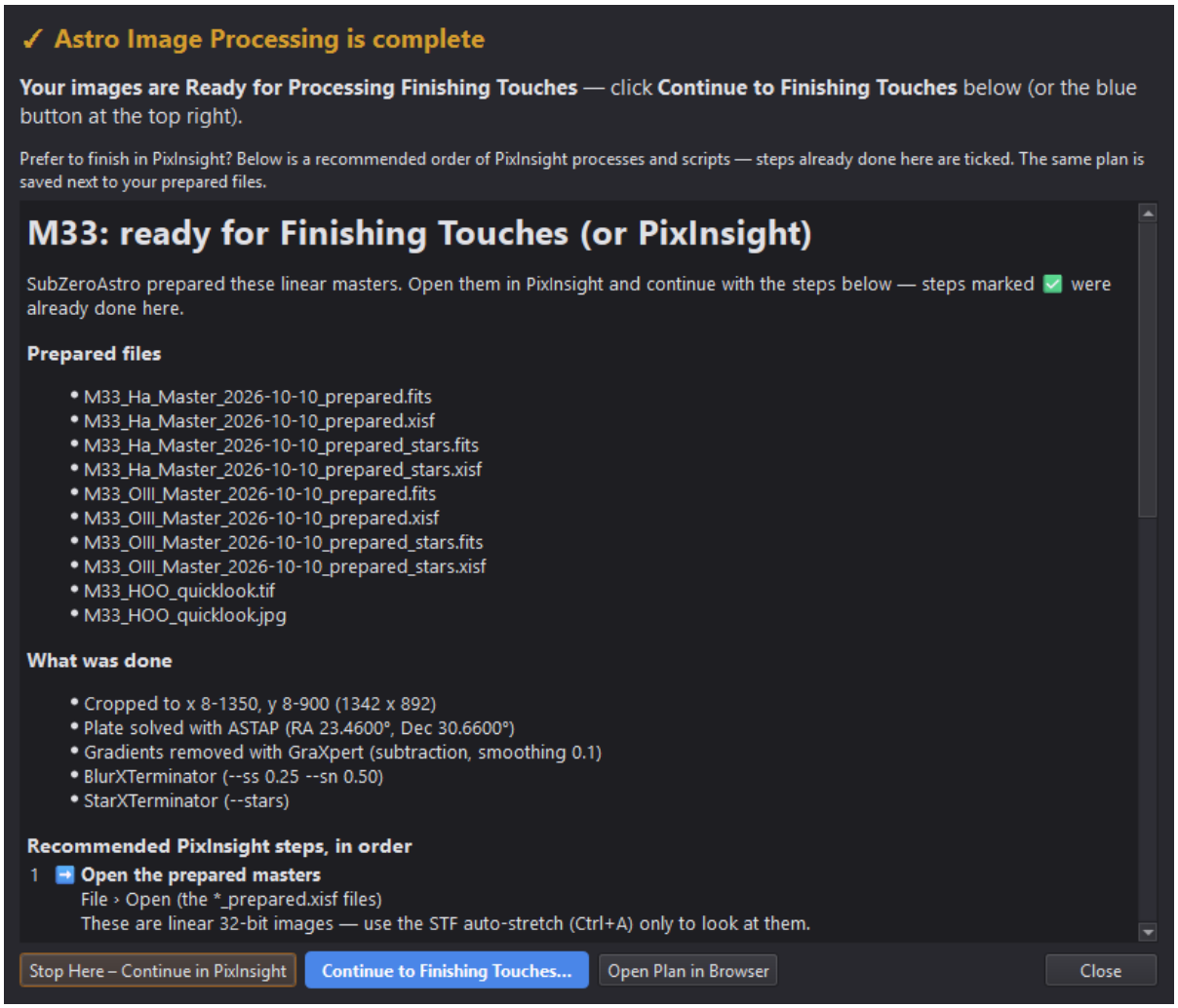
Tick the steps you want and press ▶️ **Run Selected Steps**. As each step finishes, the preview shows its result. Under **Step history**:

- **Show** - look at the result after any step.
- **Hold for Before** - press and hold to compare with the step before.
- **Go Back & Redo This Step** - change that step's settings and run again; only that step and the ones after it are redone.
- **Start Fresh** - forget the saved steps and start from the original masters.




*Step history under the preview (example data)*

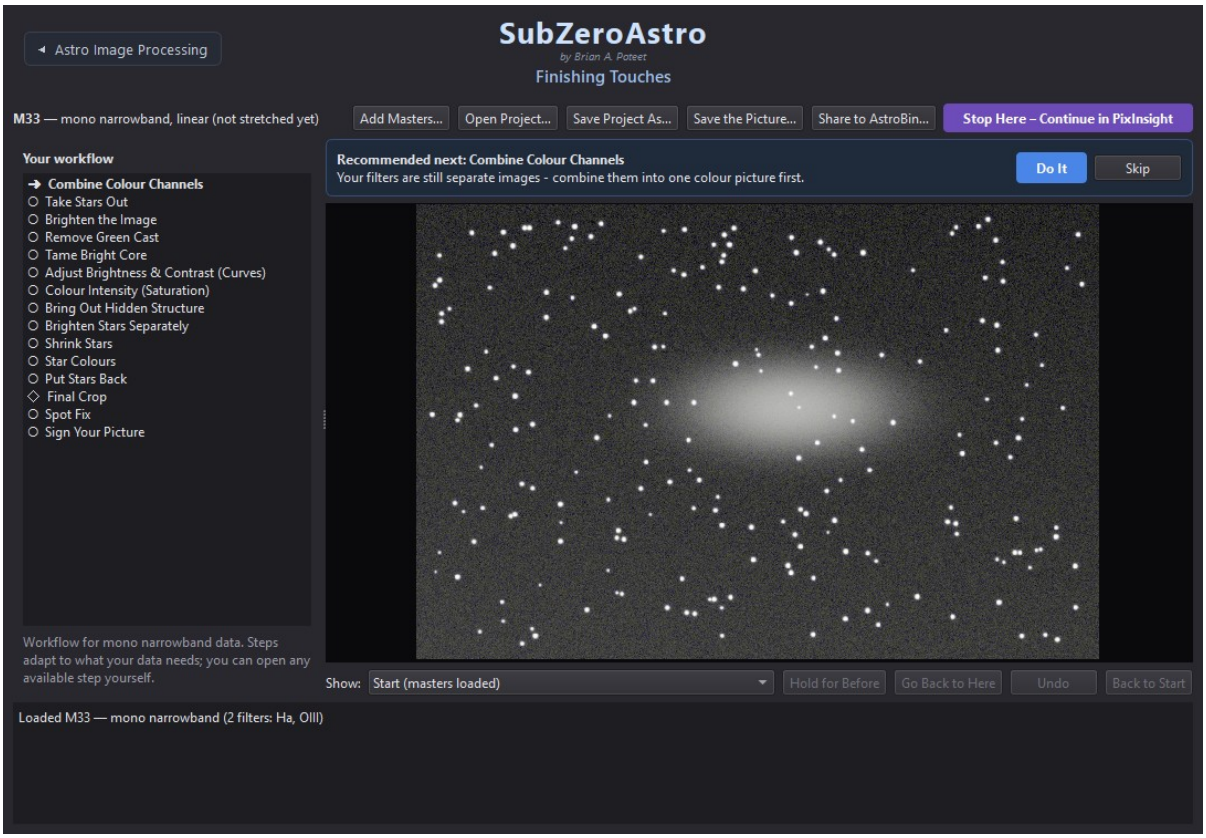
When it finishes, the bar at the bottom reads ✔ **Ready for Processing Finishing Touches**, the run button turns amber (✔ **Done – Run Again?**) and the **Astro Image Processing – Complete** window opens. It says what comes next and, for PixInsight users, lists a recommended order of PixInsight processes (also saved next to your files).




*The Astro Image Processing – Complete window*

Click **Continue to Finishing Touches ➜**, or **Stop Here – Continue in PixInsight**.

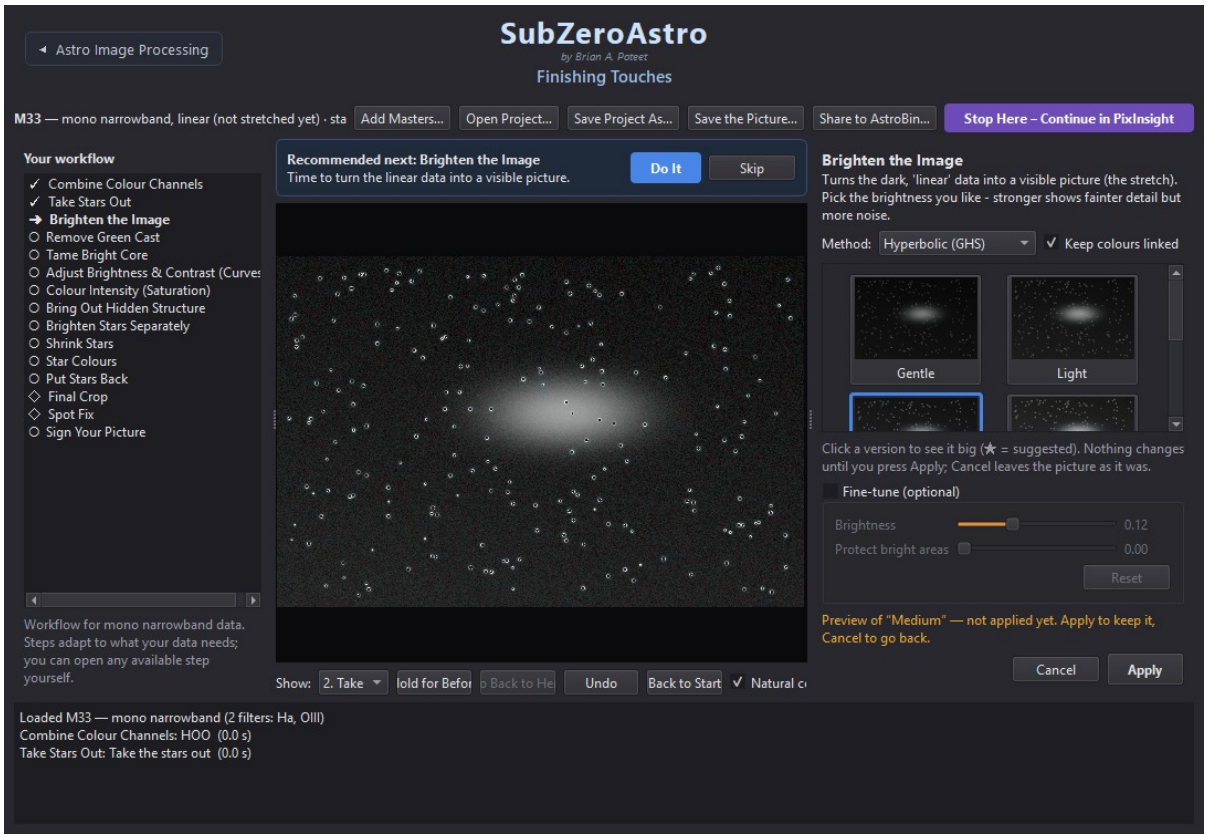
# 9 Page 4 – Finishing Touches




*Finishing Touches: the workflow (left), the recommended next step and the picture (example data)*

## 9.1 How it works

1. The **workflow list** on the left shows every step for your kind of data: ✓ done, ➜ next, ○ later, – not needed, ◇optional.
2. The card at the top says what is **Recommended next** and why. Click **Do It** (or click any step in the list).
3. Pick a version: each step shows 3-5 small pictures of **your** image - gentle to strong, or different colour mixes. ★ marks the suggestion. **Clicking only previews it big** - nothing changes yet.
4. Optional: tick **Fine-tune** and move the plain-language sliders; the preview follows. **Reset** returns to the version you picked.
5. Press **Apply** to keep it - or **Cancel** to leave the picture exactly as it was.




*Picking a version: Brighten the Image offers five strengths (example data)*

**Several masters per filter, or masters from different setups.** When you load more than one master of the same filter (other nights, other cameras or telescopes), all of them are used: they are brought to the same brightness and averaged, the less noisy ones counting more. Every master is first matched star by star to one reference master. Masters from another camera or telescope are bent slightly differently by their optics - most at the edges, which would give coloured double stars in the corners - so SubZeroAstro measures that distortion from hundreds of stars and corrects it. A master whose stars cannot be matched to the others is left out (when its filter has other masters), so it can't cause double stars, and the picture is cropped to the area all masters cover. The note under the title says how many masters were combined, corrected or left out.

## 9.2 Going back

- **Undo** (Ctrl+Z) removes the last step.
- **Show:** pick any earlier step to look at it; **Hold for Before** compares; **Go Back to Here** continues from that step.
- **Back to Start** returns to the original masters.
- Every step is saved. Opening the same masters again offers to **carry on where you left off**.

## 9.3 The steps

Which steps appear depends on your data. In typical order:

| Step | What it does | PixInsight equivalent |
|---|---|---|
| Add Hydrogen Glow | Adds Ha to the red of an RGB set, without its stars. For a galaxy (M31, M33, M101, NGC 891…) it adds only the compact HII regions - the red knots along the arms | Continuum subtraction + PixelMath / NBRGBCombination |
| Combine Colour Channels | Builds colour from separate filters: RGB, LRGB, Hubble (SHO), HSO, HOO, Foraxx… | ChannelCombination / LRGBCombination |
| Choose a Colour Palette | Dual-band colour cameras: separates Ha and OIII (or SII and OIII) and offers HOO, Gold & blue, Foraxx… | PixelMath palettes |
| Balance Colours | Neutral sky and natural star colour | BackgroundNeutralization / colour calibration |
| Remove Green Cast | Removes the green tint (or turns Hubble colours gold and blue) | SCNR |
| Mote Buster ◇ | Fix Dust Donuts: finds the round shadows of dust and divides them out with a synthetic flat - the detail underneath comes back | A new flat, or DBE with division |
| Take Stars Out | Hides the stars so the nebula can be brightened strongly (instant if page 3 already separated them) | StarXTerminator |
| Brighten the Image | The stretch - five strengths, three methods | STF / HistogramTransformation / GHS |
| Tame Bright Core ◇ | Recovers detail in a burnt-out galaxy core or bright nebula | HDRMultiscaleTransform with a mask |
| Adjust Brightness & Contrast | Curves presets, sliders and a curve you can drag | CurvesTransformation |
| Colour Intensity | Richer or calmer colour; dark sky protected | Saturation curve |
| Bring Out Hidden Structure ◇ | Local contrast for dust lanes and cores | LocalHistogramEqualization |
| Brighten Stars Separately | Gentle star stretch - small, colourful stars | Star stretch |
| Shrink Stars | Makes stars smaller | Star reduction / MorphologicalTransformation |
| Star Colours | Natural star colours (estimated from your data) or clean white stars - recommended for narrowband | Star colour correction |
| Put Stars Back | Blends the stars back at the strength you pick | PixelMath screen blend |
| Darken the Sky Background ◇ | Darker (or lighter) sky without dimming the nebula, faint dust or stars; also evens out a brighter corner or side | ABE/DBE model + PixelMath |
| Final Noise Reduction ◇ | A last, light NoiseXTerminator pass on the finished picture (needs the RC-Astro CLI) | NoiseXTerminator |

| Step | What it does | PixInsight equivalent |
|---|---|---|
| Flip the Picture ◇ | Flip Horizontally and/or Flip Vertically (each press toggles) - the saved pictures are flipped too | FastRotation |
| Final Crop ◇ | A last clean-up crop; finds ragged or off-colour edges for you | DynamicCrop |
| Spot Fix ◇ | Click dust donuts, satellite remnants or hot pixels to patch them | CloneStamp |
| Sign Your Picture ◇ | Your signature - and the target name above it - in a corner or wherever you drag it | Annotation |

◇= optional: offered in the list but not pushed by *Recommended next*.

## 9.4 Mote Buster: dust donuts

Dust on a filter or on the camera window casts round shadows - **dust donuts** (rings with a reflector, discs with a refractor). A good flat removes them; when the flat is missing or no longer matches, **Mote Buster – Fix Dust Donuts** helps. A dust shadow does not wipe anything out: it lets through the same fraction of light everywhere inside it, nebula included. So SubZeroAstro measures each donut's shadow ring by ring - in 16 directions against the sky just outside it, levelled for any gradient - builds a **synthetic flat** just for that donut and divides it out, the way a uniformity (flood-field) correction works. The detail underneath comes back instead of being painted over.

- Open **Mote Buster** while the picture is still unstretched (it comes before *Take Stars Out*). The donuts it finds in the sky are already circled in yellow.
- Donuts on bright nebulosity are not searched automatically (a gap between filaments could look like one) - **click** those yourself. Click to place a circle, then **turn the mouse wheel to size that circle** while you watch (just cover the donut). Click a circle to pick it again (it turns blue) and drag to move it; right-click removes it. A dashed ring follows the mouse to show the size of the next circle. Clicks need not be exact: each circle is centred on its donut automatically.
- Check the preview, then **Apply**. For small specks (hot pixels, satellite remnants) use **Spot Fix** instead.

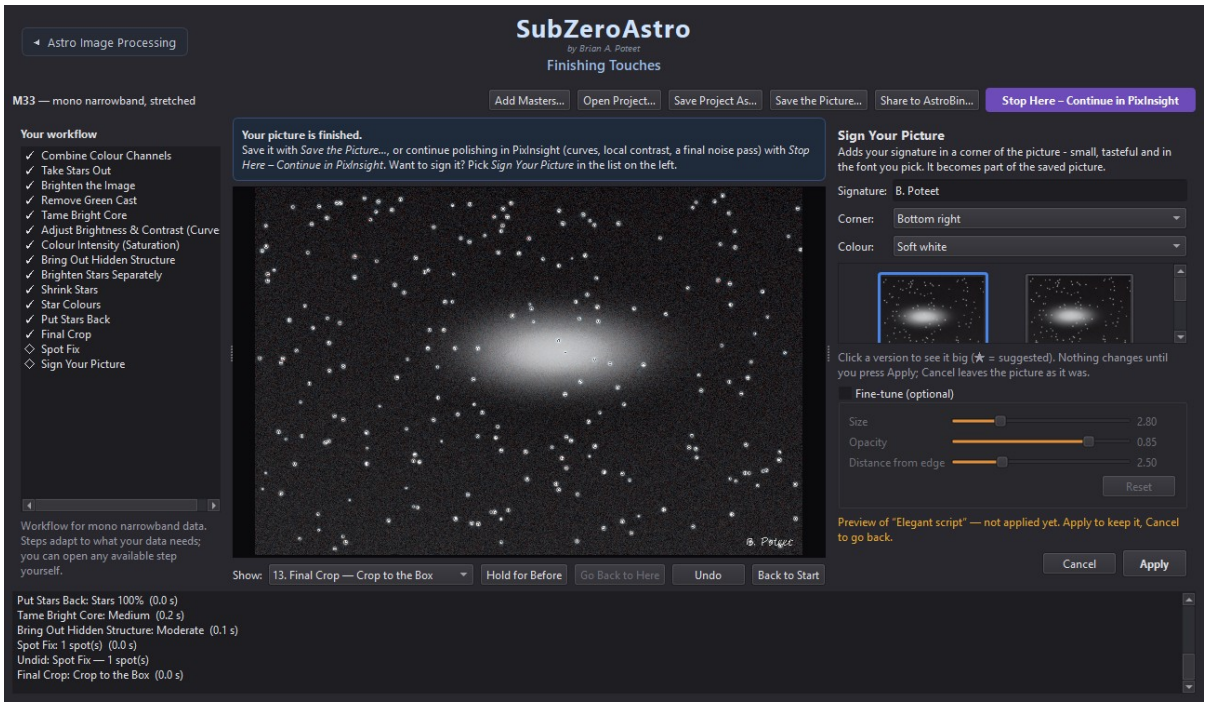
## 9.5 Stars: take out, brighten, colour, put back

Stars and nebula look best when processed separately. **Take Stars Out** hides the stars (the picture shows the nebula alone); the nebula steps follow; then **Brighten Stars Separately**, **Shrink Stars** and **Star Colours** work on the stars (their previews show the stars); **Put Stars Back** recombines them. If you **Skip** Take Stars Out, the stars simply stay in the picture.

## 9.6 Darker sky, Final Crop, Spot Fix and your signature

- **Darken the Sky Background** is for a sky that looks grey or washed out. Pick *Slightly darker*, *Darker*, *Much darker* or *Lighter* (each shows the sky's brightness before and after), then fine-tune with **Sky darkness**. It works like PixInsight's background extraction: SubZeroAstro maps the sky glow across the picture (a nebula cannot push that map up) and takes part of it away. Everything that shines above the sky - nebula, faint dust, stars - keeps exactly the brightness it had **above the sky**, so faint filaments stay visible and stars are not dimmed. A brighter corner or side loses more, so the sky also evens out. Darker skies show grain more: a lighter setting often looks more natural.

- **Final Noise Reduction** runs NoiseXTerminator once more on the finished picture - most useful right after a darker sky, which shows grain more. The choices (*Light*, *Medium*, *Strong*) are shown at full size on the busiest part of the picture so you can judge the grain; **Strength** fine-tunes it. (BlurXTerminator is not offered here on purpose: it is made for unstretched data and already runs on page 3. For more crispness at the end, use **Bring Out Hidden Structure**.)
- **Final Crop** shows a yellow box on the whole picture, already placed where it found ragged or off-colour edges. Drag it or its sides (**Found Edges** / **Whole Picture** reset it) and press **Crop to the Box**.
- **Spot Fix**: click the middle of a blemish - a circle appears - then **turn the mouse wheel to make that circle bigger or smaller** until it just covers the blemish. Click a circle to pick it again (it turns blue) and drag to move it; right-click removes it. A dashed ring follows the mouse to show the size of the next circle. Then **Fix the marked spots**.
- **Flip the Picture** (optional): press **Flip Horizontally** (left ↔ right) and/or **Flip Vertically** (upside down) - each press flips the preview, pressing again flips it back (both together turn it 180°). When it looks right, press **Keep This Orientation**. Useful to match how the object usually looks in charts and other pictures. Flip before signing, so the signature reads the right way.
- **Sign Your Picture**: type your name once (it is remembered) and, if you like, a **Target name** for a line above it - filled in for you (for example *NGC 281 "Pacman Nebula"*) and remembered for each target; empty it for none. Pick a font - *Elegant script*, *Handwritten* or *Classic* - a corner and a colour, and fine-tune size, opacity and distance from the edge. To put it anywhere else, **drag the dashed box** on the picture (or click where it should go); *Place* then says *Where I dragged it*. Do this last: the signature becomes part of the picture.




*Sign Your Picture: signature text, corner, colour and three fonts (example data)*

## 9.7 Saving and sharing

- **Save the Picture…** - TIFF (16-bit), JPG, PNG, XISF or FITS. All the files of one picture go into **their own folder inside the one you pick**, named after the target and telescope - for example *Final Astro Pics/M101_RedCat51/* holds M101_finished.tif, .jpg, .fits and the description. The folder you pick is remembered: each target keeps its own, and a new target starts in the folder

you used last. Tick **Also save a short description of the target** to save *<target>_about.txt* next to the pictures: the opening paragraphs of its Wikipedia article plus **facts from SIMBAD** (the professional astronomy database at CDS, Strasbourg) - the kind of object, its position (RA / Dec) and brightness. Faint objects without a Wikipedia article still get the SIMBAD facts. It is shown first so you can edit it (or type your own when there is no internet), and both sources are credited.

- **Save Project As… / Open Project…** - keep every step in a folder of your choice and carry on later.
- **Share to AstroBin…** - saves an upload-ready JPG (and optional TIFF) plus an **acquisition CSV** with every night's subs, exposure, gain, temperature and FWHM, then opens AstroBin's upload page. Upload the picture, and on the *Acquisition* step choose **Import CSV**.
- **Stop Here – Continue in PixInsight** - see 5.4.

## 9.8 Starting your next target

When a picture is finished, click the teal ↻ **End this Project and Start New Project** button at the top right of Finishing Touches. SubZeroAstro asks first - and if you have not saved the picture yet, offers **Save the Picture First…**. Then every page and image closes, the helper programs are checked again, and you are back on **page 1 (Quality and Cull)** with the **Load Light Frame Data** window open, ready for the next target's light frames: cull, stack, process and finish, from the beginning.

> **Your work is safe:** Nothing on your disk is deleted. Your masters, finished pictures and every Finishing Touches step stay where they are: open the same masters on Finishing Touches later and the project carries on where you stopped.

# 10 Recipes for common setups

## 10.1 Colour camera (OSC), broadband

Cull › Stack (flats + darks or bias) › page 3: crop, plate solve, GraXpert, BlurX, StarX › page 4: Balance Colours, Remove Green Cast, Take Stars Out, Brighten, Curves, Colour Intensity, star steps, Put Stars Back, Final Crop, Sign.

## 10.2 Colour camera with a dual-band filter

Make sure the filter is set (page 1, *Filter used* or *Set Filter Name*) so the matching flats are found. On page 4 the first step is **Choose a Colour Palette** (HOO, Gold & blue, Foraxx…), and **Star Colours** is recommended so the stars are not all teal. Two dual-band filters of the same target - **L-Ultimate + L-Synergy** or **Askar D1 + D2** - are combined into a full **SHO** set automatically.

## 10.3 Mono camera: LRGB

**RGB (or LRGB) + Ha of a galaxy**: SubZeroAstro recognises galaxies by name (M31, M33, M51, M81, M101, NGC 891, IC 342… or names like *Andromeda* or *Pinwheel*). **Add Hydrogen Glow** then recommends **HII regions**: the starlight - and the galaxy's own smooth light - is subtracted from the Ha, and only the compact hydrogen knots that clearly stand out are added to red. They stand out crisply against the blue arms without tinting the whole galaxy pink. Tick **Show the hydrogen regions** to see exactly what is added; *All hydrogen glow* adds everything, for galaxies set inside nebulosity.

Each filter is stacked separately on page 2 (one row each; keep *Line up all filters* on). Load all masters together on page 3 - they must appear as **one row**. Page 4 starts with **Combine Colour Channels** (RGB or LRGB).

## 10.4 Mono camera: narrowband (SHO)

As 10.3 with Ha, OIII and SII. Filter names from filter wheels such as *Ha 7nm*, *H-alpha*, *O*, *S* or *Antlia Ha 3nm Pro* are recognised. Page 4: **Combine Colour Channels** › Hubble (SHO), Foraxx or HOO; then Remove Green Cast (turns the green gold), and **Star Colours** before Put Stars Back.

## 10.5 Smart telescopes (Seestar, DWARF, Vespera)

Raw subs are recognised as lights automatically, with no calibration needed (the telescope calibrates on board). Alt-az field rotation is handled during alignment and the rotated edges are cropped. Seestar *Stacked_…* files from several nights can be combined on page 2, or a single stack can go straight to page 3.

## 10.6 Comets

A comet moves against the stars while you image it, so a normal stack (lined up on the stars) smears it, and a stack lined up on the comet trails the stars. SubZeroAstro does both and joins them. It recognises a comet by its name (*C/2023 A3*, *12P/Pons-Brooks*, *161P Hartley-IRAS*, or any name containing *comet*) - page 2's plan shows  for it. While stacking it finds the comet's head in every frame, works out its motion from the frames' times, makes a **second stack that follows the comet** (the stars, which move through it, are rejected), and saves three masters: the main one with **sharp stars and a sharp comet**, plus **_Comet** (comet only) and **_Stars** (star-aligned) next to it. The log says how far the comet moved.

When the comet barely moved during the session (under 3 pixels - common with short sessions), or it is too faint or too uncertain to follow, only the ordinary stack is saved and the log says why. Long sessions of several frames per minute work best. On page 4 the main master goes through the usual steps; for a comet **Darken the Sky** is recommended, while star steps (*Take Stars Out*, *Shrink Stars*) are optional.

## 10.7 Target types

From the target's name SubZeroAstro works out what kind of object it is - **galaxy, nebula, planetary nebula, star cluster or comet** - and shows it in the title of page 4. The type only fine-tunes the recommendations; **no step is ever taken away**. Galaxies get *Local Contrast* recommended and HII regions in *Add Hydrogen Glow*; star clusters make star removal and shrinking optional (the stars are the subject); comets get a comet stack on page 2. When the name gives nothing away, the usual workflow for your data applies.

# 11 Troubleshooting and questions

## 11.1 Common questions

| Problem | What to do |
|---|---|
| Windows: "Windows protected your PC" | Click **More info › Run anyway** (3.1). |
| Mac: "can't be opened" or "is damaged" | Right-click › Open, or the Terminal command in 3.2. |
| Frames show **No filter** | Set the filter on page 1 (*Filter used* or *Frame › Set Filter Name…*). Needed to match flats and to offer the right palettes. |
| Flats column is ✗ red | No flats for that filter/rig. Change the row's Filter to match your flats, or add the flats folder. |
| Bright corners and dark rings | Flats calibrated without bias/dark-flats, or flats that don't match. Check the ⚠️ note on page 2 and add a bias or dark-flats (7.2). |
| Filters of one target in separate rows on page 3 | Check that the target names match (OBJECT). Masters stacked in separate runs are lined up automatically. |
| No *Combine Colour Channels* on page 4 | The masters were not loaded together, or a filter name was not recognised - the log lists the filters it found. |
| Plate solve failed | SubZeroAstro already retries with other field sizes (wrong binning or a reducer in the header are the usual cause; when the header has no focal length at all - Telescope Live files say 0 - typical fields from 0.25° to 4.5° are tried, and the one that works is reused for the other masters). If it still fails, read the log line: for small fields (under about 0.5°, e.g. galaxies with long focal lengths) ASTAP needs the **D50** or **D80** star database; with no pointing in the header tick **Search the whole sky**. A failed solve only means the picture has no sky coordinates - every other step works as usual. |
| Tools show *not found* | Click **Browse…** and point at the program, or use **Settings › Your Tools…** (4.1). |
| Amber banner on page 1 | A helper program is missing or not licensed. **Set Up…** fixes it; **Don't Remind Me** if you don't use it (4.1). |
| An XTerminator says *not licensed* | Click **RC-Astro Licences…** on page 3 and activate with the code from your rc-astro.com account (4.1). |
| Frames skipped: *could not match the star field* | Usually clouds, trailing or a badly out-of-focus frame. Frames saved mirror-image (some capture programs do this) are recognised and lined up automatically. |
| XTerminators very slow | Choose your graphics card under **RC-Astro device** (4.1). |
| Slow on a network drive | Use *Star measurement: Fast*; hide the image (Ctrl+I); results are cached for next time. |
| Something went wrong mid-run | Restart: SubZeroAstro offers to resume from its saved progress. |

| Problem | What to do |
|---|---|
| Half the frames show *unreadable*, names starting with `._` | Those are hidden macOS copies (AppleDouble files) that a Mac writes next to every file on a NAS or external drive - not pictures. Version 0.9.31 and later ignore them; with an older version, delete the `._` files or turn off *Include subfolders*. |

## 11.2 Getting help and reporting problems

**Zip My Logs**: the button at the top left of every page (also **Help › Zip My Logs**) puts the log of every open page, the stack logs of your masters, page 3's processing report and a short note about your computer (SubZeroAstro version, system, screen, which helper programs were found) into one zip on your Desktop and shows it - text only, no pictures. Attach that zip to your message: it is the quickest way to get help.

Note the version (in small print under the name at the top of every page, or **Help › About**) and take a screenshot of the page and its log. Problems and ideas can be reported on the GitHub page (**github.com/bapoteet/SubZeroAstro**, *Issues*). Advanced: `SubZeroAstro --selftest` runs the built-in self-test.

# 12 Keyboard shortcuts

## 12.1 Page 1 – Quality and Cull

| Keys | Action |
|---|---|
| ↑ ↓ ← → | Previous / next frame |
| Shift + arrows, Shift / Ctrl + click | Select several frames |
| X or Delete | Reject / restore the selected frame(s) |
| F | Flag / unflag |
| Shift+X | Reject all flagged frames |
| D | Clear all flags |
| Ctrl+F | Auto-Flag… |
| Ctrl+Z | Undo |
| Ctrl+0 / 1 / double-click | Fit to window / 100% / toggle |
| Mouse wheel, drag | Zoom, pan |
| Ctrl+O | Load Light Frame Data |
| Ctrl+Shift+O | Open one folder |
| Ctrl+N | Start over – load new data |
| Ctrl+H | FITS header window |
| Ctrl+I | Show / hide the image |
| Ctrl+E | Save a text list of kept frames |
| F5 | Reload folder |

## 12.2 Page 4 – Finishing Touches

| Keys | Action |
|---|---|
| Ctrl+Z | Undo the last step |
| Mouse wheel (Spot Fix) | Spot size |
| Right-click (Spot Fix) | Remove the last spot |

# 13 Files and folders SubZeroAstro creates

## 13.1 Next to your images

| Where | What |
|---|---|
| `<lights folder>\REJECTED\` | Rejected sub-frames (moved, never deleted) |
| `<calibration folder>\<rig>\ Masters\…` | Calibration masters (bias, darks, flats per filter) |
| `<master lights folder>\<target>\` | Master lights (`…_Master_<dates>.fits/.xisf`) and their `_stacklog.txt` |
| `Prepared\` (or your chosen folder)\ `<target>\` | Page 3 results: `…_prepared.xisf`, `… _prepared_stars.xisf`, quick-look JPG/TIFF, PixInsight next steps |
| `Finished\`, `PixInsight\`, `AstroBin\` | Page 4 saves, PixInsight hand-overs and AstroBin uploads |

## 13.2 On your computer

| What | Windows | Mac |
|---|---|---|
| Saved progress | `%LOCALAPPDATA%\SubZeroAstro\ checkpoints` | `~/Library/Application Support/SubZeroAstro` |
| Measurement cache | `%LOCALAPPDATA%\ FITSBlasterWin\cache` | `~/Library/Caches/SubZeroAstro` |
| Settings | Windows registry (SubZeroAstro settings) | `~/Library/Preferences` |

# 14 Privacy, licence and credits

## 14.1 Privacy

SubZeroAstro does not send any information anywhere unless you ask it to. It reads and writes images only where you point it. **Share to AstroBin…** only opens AstroBin's page in your browser. Companion programs (ASTAP, GraXpert, RC-Astro) are separate programs with their own terms.

## 14.2 Licence and credits

SubZeroAstro is free software under the **MIT License**. Created by **Brian A. Poteet**, built with Claude (Anthropic). The culling view was inspired by FITS Blaster for macOS by Tom van Peer. PixInsight, BlurXTerminator, NoiseXTerminator, StarXTerminator, GraXpert, ASTAP, AstroBin and the telescope and filter names mentioned are trademarks of their respective owners.

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